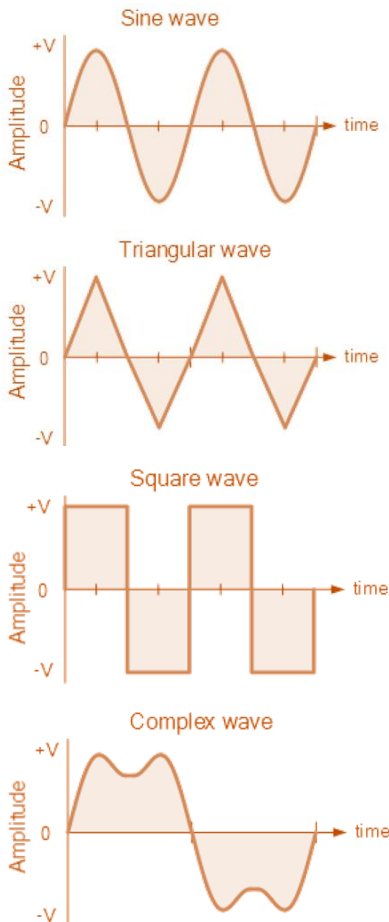




Alternating Current

Alternating Current (AC) is defined as one that continuously varies in amplitude and periodically reverses in polarity.

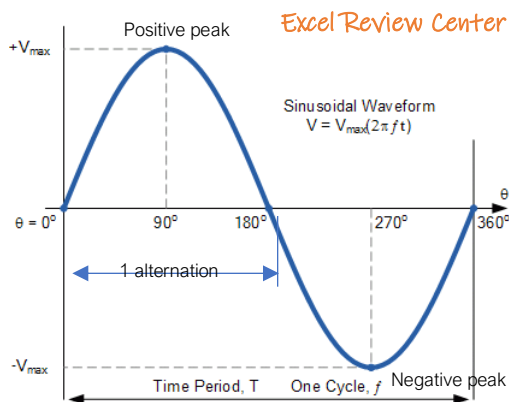
Examples of alternating waveforms:



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The Sine Wave

The **sine wave** is the standard ac waveform.

**Note:**

This waveform is called a sine wave, sinusoidal wave or sinusoid because the amount of induced voltage is proportional to the sine of the angle of rotation in the circular motion producing the voltage.

TERMS relating to AC

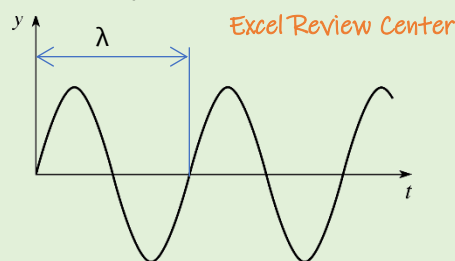
Frequency (f) is the number of cycles per second in Hertz (Hz) units.

$$1 \text{ cycle/second} = 1 \text{ Hz}$$

Period (T) is the amount of time for one cycle. The frequency and period are reciprocals of each other.

$$T = \frac{1}{f}$$

Wavelength (λ) is the length of one complete wave or cycle.



The wavelength (λ) depends upon the frequency of the variation and its velocity of transmission:

$$\lambda = \frac{\text{velocity}}{\text{frequency}}$$

Values of Sine Wave

Instantaneous value is the value of a sine wave at any instant of time. It is denoted by lowercase letters v or i for voltage and current.

$$v = V_m \sin 2\pi ft = V_m \sin \omega t = V_m \sin \theta$$

$$i = I_m \sin 2\pi ft = I_m \sin \omega t = I_m \sin \theta$$

where:

θ = the angle of rotation (rad)

ω = the angular velocity of a radius vector generating the waveform (rad/sec)

t = time in seconds

V_m = maximum voltage value

I_m = maximum current value

Peak value is the maximum value represented by V_m or V_p for voltage and I_m or I_p for current.

$$V_m \text{ or } V_p = \sqrt{2} V_{rms} = 1.414 V_{rms}$$

$$I_m \text{ or } I_p = \sqrt{2} I_{rms} = 1.414 I_{rms}$$

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Peak-to-peak value is the value from positive peak to the negative peak represented by V_{p-p} for voltage and I_{p-p} for current.

$$V_{p-p} = 2V_p = 2.828V_{rms}$$

$$I_{p-p} = 2I_p = 2.828I_{rms}$$

Average value is the arithmetic average of all the values in a sine wave for one alternation, or a half-cycle.

$$V_{avg} = \frac{2V_m}{\pi} = 0.637V_m$$

$$I_{avg} = \frac{2I_m}{\pi} = 0.637I_m$$

Root-Mean-Square (RMS) or Effective value (V_{eff} or I_{eff}) is the value of a sine wave at 45° which is 70.7% of the peak value.

RMS value = Peak value x sin 45°

$$V_{rms} = 0.707V_p = \frac{V_p}{\sqrt{2}}$$

$$I_{rms} = 0.707I_p = \frac{I_p}{\sqrt{2}}$$

Note:

An alternating voltage with an rms value of 120 V, for instance, is just as effective in heating a filament of a light bulb as 120 V from a steady dc voltage source. For this reason, the rms value is also the effective value.

Form & Peak Factors

Form factor is the ratio of the rms value to the average value.

Peak factor is the ratio of peak value to the rms value.

For a sine wave,

$$\text{Form factor} = \frac{V_{rms}}{V_{avg}} = \frac{0.707V_m}{0.637V_m} = 1.11$$

$$\text{Peak factor} = \frac{V_m}{V_{rms}} = \frac{V_m}{0.707V_m} = 1.4142$$



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